

A Computational Framework for the Composite Action of Fundamental Physics, with a Reproduction of Gauge-Coupling Unification

Professor Codephreak*

June 2026

Abstract

We present a pedagogical and computational framework that organizes the standard action functionals of fundamental physics — the Einstein–Hilbert action, the Dirac and Higgs actions in curved spacetime, the Yang–Mills gauge action, and the formal loop expansion — into a single composite expression, and computes the piece of unification physics that is honestly computable at this level: the renormalization-group running of the three gauge couplings at one and two loops, the resulting unification scale in the minimal supersymmetric extension, and the proton-lifetime estimate it implies. The composite action is presented as a *definition* (an organizing framework), not a theorem of unification: it is an additive juxtaposition of separately established actions and carries no unifying dynamics. The quantitative content is a reproduction of known results: the Standard Model couplings miss a common intersection by 11–13%, while the MSSM couplings unify to 0.7% (two loops) at $M_{\text{GUT}} \approx 1.3 \times 10^{16}$ GeV with $\alpha_{\text{GUT}}^{-1} \approx 25$, yielding $\tau_p \sim 10^{35-36}$ yr — above the Super-Kamiokande bound — whereas the Standard Model near-miss scale is excluded by four orders of magnitude. We state precisely what would be required for a framework of this kind to rise from organized compilation to scientific theory, and we make every number reproducible by running the accompanying code.

1 Introduction

A theory of everything — a single, internally consistent framework containing the four fundamental interactions and all matter — remains one of the major unsolved problems in physics. The principal research programs (string/M-theory, loop quantum gravity, asymptotic safety) are mathematically demanding, mutually incompatible in their foundations, and, to date, experimentally unconfirmed.

This project began as an AI-assisted exploration that assembled the standard actions of fundamental physics into a composite expression and visualized them. An independent academic evaluation [19] examined that corpus and found it to be a faithful *compilation* of textbook actions but not a theory: it contained no formal apparatus, no derivations, an internally inconsistent gravity sector, and no falsifiable output. The present document is the response: the same material reorganized with the formal apparatus the evaluation found missing, the internal

*An AI-assisted project of Gregory L. Magnusson. Restructured in 2026 in response to an independent academic evaluation [19]; code and documentation at <https://github.com/GrandUnifiedTheoryOfEverything/gutoe>.

inconsistencies repaired by honest reframing, and — most importantly — the addition of the unification computation that any work in this area must exhibit (Section 5).

Nothing in this paper is novel physics. The contribution is structural and pedagogical: a single, runnable repository in which the assembled inputs to the unification problem, the parts that can be computed, and the precise gap to a real theory are presented together.

2 Conventions

Metric signature $(-, +, +, +)$; spacetime dimension four unless stated; natural units $\hbar = c = 1$ in field-theoretic expressions. Hypercharge is normalized by $Q = T_3 + Y$, and the GUT-normalized $U(1)$ coupling is $\alpha_1 = \frac{5}{3}\alpha_Y$, a factor *derived* in Section 4. Physical constants [13]: $G = 6.674\,30 \times 10^{-11} \text{ m}^3 \text{ kg}^{-1} \text{ s}^{-2}$, $\hbar = 1.054\,571\,817 \times 10^{-34} \text{ J s}$, and the cosmological constant

$$\Lambda = 1.1056 \times 10^{-52} \text{ m}^{-2} \quad (1)$$

from Planck 2018 [16]. We record the units explicitly: a dimensionless value for Λ is meaningless, an error of the pre-2026 corpus corrected here.

3 The composite action

Definition 1 (Composite action; organizing framework).

$$S = S_{\text{gravity}} + S_{\text{matter}} + S_{\text{gauge}} + S_{\text{quantum}}, \quad (2)$$

$$S_{\text{gravity}} = \frac{1}{16\pi G} \int d^4x \sqrt{-g} (R - 2\Lambda), \quad (3)$$

$$S_{\text{matter}} = \int d^4x \sqrt{-g} \left[\bar{\psi} (i\gamma^\mu D_\mu - m) \psi + (D_\mu \phi)^\dagger (D^\mu \phi) - V(\phi) \right], \quad (4)$$

$$S_{\text{gauge}} = -\frac{1}{4} \int d^4x \sqrt{-g} F_{\mu\nu}^a F^{a\mu\nu}, \quad (5)$$

$$S_{\text{quantum}} = \sum_{n=1}^{\infty} \hbar^n S_n, \quad (6)$$

with $V(\phi) = -\mu^2|\phi|^2 + \lambda|\phi|^4$ ($\mu^2, \lambda > 0$) and $F_{\mu\nu}^a = \partial_\mu A_\nu^a - \partial_\nu A_\mu^a + gf^{abc}A_\mu^b A_\nu^c$.

Definition 1 *organizes* the standard actions of fundamental physics in one expression. It is a definition, not a theorem: the sum carries no unifying dynamics beyond that of its parts. Each term is a transcription of an established, separately tested action; what a genuine unification requires — a single gauge group from which the forces emerge, derived inter-sector couplings, a consistent quantization — is exactly what the bare sum does not supply. We return to this distinction in Section 7.

Remark 1 (The gravity sector: one action, several research programs). Only the Einstein–Hilbert action enters S . The loop-quantum-gravity (Ashtekar-variable) action and the string-theoretic (NS–NS sector) effective action, which earlier versions of this project listed as parallel “formulations” of S_{gravity} , are mutually exclusive candidate UV completions: they live in different dimensions (4 vs. 10), are built on different fundamental variables, and belong to competing research programs [11, 12]. They may be compared; they may not be summed. A unified theory must select or derive one and recover the others’ regimes as limits — a derivation that exists neither here nor, to date, anywhere.

4 Embedding the Standard Model in SU(5) and SO(10)

This section is an exposition of known results [1, 3], included because it is the group-theoretic arithmetic any unification claim must exhibit, and because the RG analysis of Section 5 requires the normalization derived here. Every number is reproduced by `python3 -m toe_math.gut_embedding`.

One generation of Standard Model fermions (15 left-handed Weyl spinors) fits exactly into $\bar{\mathbf{5}} \oplus \mathbf{10}$ of SU(5) (Table 1).

Field	$SU(3)_c$	$SU(2)_L$	Y	States	$SU(5)$ irrep
$Q = (u_L, d_L)$	$\mathbf{3}$	$\mathbf{2}$	$1/6$	6	$\mathbf{10}$
u_L^c	$\bar{\mathbf{3}}$	$\mathbf{1}$	$-2/3$	3	$\mathbf{10}$
e_L^c	$\mathbf{1}$	$\mathbf{1}$	1	1	$\mathbf{10}$
d_L^c	$\bar{\mathbf{3}}$	$\mathbf{1}$	$1/3$	3	$\bar{\mathbf{5}}$
$L = (\nu_L, e_L)$	$\mathbf{1}$	$\mathbf{2}$	$-1/2$	2	$\bar{\mathbf{5}}$

Table 1: One SM generation in SU(5). Conventions: left-handed Weyl fermions, $Q = T_3 + Y$.

The hypercharge normalization, derived. In SU(5) all generators are normalized identically, so $\text{Tr}(T^2)$ over a generation must agree between T_3 and the properly normalized hypercharge generator. From Table 1,

$$\text{Tr}(Y^2) = 6 \cdot \frac{1}{36} + 3 \cdot \frac{4}{9} + 1 + 3 \cdot \frac{1}{9} + 2 \cdot \frac{1}{4} = \frac{10}{3}, \quad \text{Tr}(T_3^2) = 2, \quad (7)$$

hence $T_1 = \sqrt{3/5} Y$ and $\alpha_1 = \frac{5}{3} \alpha_Y$.

Anomaly cancellation. Within one generation the four gauge anomaly sums vanish exactly: $\sum Y = 0$, $\sum Y^3 = 0$, $\sum_{\text{colored}} Y = 0$ ($SU(3)^2$ - $U(1)$), and $\sum_{\text{doublets}} Y = 0$ ($SU(2)^2$ - $U(1)$).

SO(10). The spinorial $\mathbf{16}$ branches under SU(5) as $\mathbf{16} = \mathbf{10} \oplus \bar{\mathbf{5}} \oplus \mathbf{1}$, accommodating one generation plus a right-handed neutrino — the natural seed of the seesaw mechanism [3].

Proposition 1 (Embedding arithmetic). *One SM generation fits exactly into $\bar{\mathbf{5}} \oplus \mathbf{10}$ of SU(5); the normalization $\alpha_1 = \frac{5}{3} \alpha_Y$ follows from $\text{Tr}(Y^2) = 10/3$; all four anomaly sums vanish; and $\mathbf{16} = \mathbf{10} \oplus \bar{\mathbf{5}} \oplus \mathbf{1}$ under $SO(10) \supset SU(5)$.* Proof: by computation (`toe_math/gut_embedding.py`).

This arithmetic shows the Standard Model *fits* in SU(5)/SO(10) — a necessary condition and the reason GUTs are taken seriously. It does not establish unification (Section 7).

5 Renormalization-group analysis

Every number in this section is reproduced by `python3 -m toe_math.rg_running`.

5.1 Inputs and boundary conditions

PDG 2024 world averages in the $\overline{\text{MS}}$ scheme at $\mu = M_Z = 91.1876 \text{ GeV}$ [13]: $\alpha_{em}^{-1}(M_Z) = 127.951$, $\sin^2 \theta_W(M_Z) = 0.23122$, $\alpha_s(M_Z) = 0.1179$. In GUT normalization,

$$\alpha_1^{-1} = \frac{3}{5} \cos^2 \theta_W \alpha_{em}^{-1} = 59.02, \quad \alpha_2^{-1} = \sin^2 \theta_W \alpha_{em}^{-1} = 29.58, \quad \alpha_3^{-1} = \alpha_s^{-1} = 8.48. \quad (8)$$

5.2 Running

At one loop, $\alpha_i^{-1}(\mu) = \alpha_i^{-1}(M_Z) - \frac{b_i}{2\pi} \ln \frac{\mu}{M_Z}$ with

$$b^{\text{SM}} = \left(\frac{41}{10}, -\frac{19}{6}, -7\right), \quad b^{\text{MSSM}} = \left(\frac{33}{5}, 1, -3\right). \quad (9)$$

At two loops the coupled system

$$\frac{d\alpha_i^{-1}}{d \ln \mu} = -\frac{1}{2\pi} \left[b_i + \frac{1}{4\pi} \sum_j b_{ij} \alpha_j \right] \quad (10)$$

is integrated numerically with the standard matrices b_{ij} [8, 6]; Yukawa contributions are neglected (a percent-level effect dominated by the top Yukawa). For the MSSM, SM beta functions are used from M_Z to the superpartner threshold m_{SUSY} (1 TeV unless stated) and MSSM beta functions above it.

5.3 Unification criterion and results

Pairwise crossing *scales* spread widely even for good unification, because the running lines intersect at shallow angles; the honest discriminator is the *coupling-space mismatch*: the relative deviation of α_3^{-1} from $\alpha_1^{-1} = \alpha_2^{-1}$ at their crossing. We call the couplings unified when the mismatch is below 3%.

Two-loop running of the gauge couplings, GUT normalization

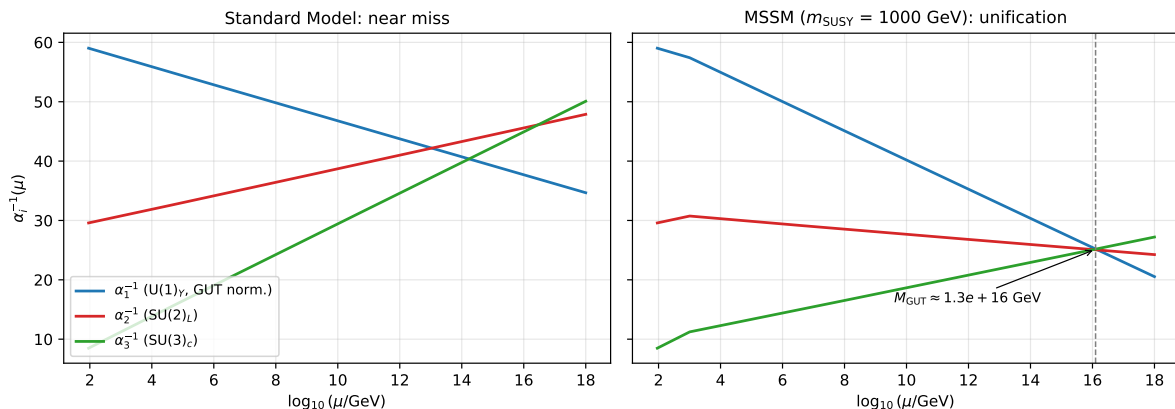


Figure 1: Two-loop running of the GUT-normalized inverse couplings. Left: Standard Model — the couplings approach but miss a common point. Right: MSSM with $m_{\text{SUSY}} = 1 \text{ TeV}$ — unification at $M_{\text{GUT}} \approx 1.3 \times 10^{16} \text{ GeV}$. Generated by `toe_math/rg_running.py`.

Proposition 2 (Standard Model near-miss). *With the stated inputs, the SM couplings fail to unify: at the $\alpha_1 = \alpha_2$ crossing, α_3^{-1} deviates by 13.1% (one loop) and 11.5% (two loops) — an order of magnitude beyond experimental uncertainty. Proof: by computation.*

Proposition 3 (MSSM unification). *With $m_{\text{SUSY}} = 1 \text{ TeV}$, the couplings unify to 2.3% (one loop) and 0.7% (two loops) at $M_{\text{GUT}} \approx 1.3\text{--}1.5 \times 10^{16} \text{ GeV}$ with $\alpha_{\text{GUT}}^{-1} \approx 25$, reproducing [5, 6]. Proof: by computation.*

Model	Loops	M_{GUT} [GeV]	α_{GUT}^{-1}	Mismatch	τ_p [yr]
SM	1	6.2×10^{14}	41.9	13.1%	7.5×10^{30}
MSSM	1	1.5×10^{16}	25.7	2.3%	1.0×10^{36}
SM	2	3.8×10^{14}	41.7	11.5%	1.0×10^{30}
MSSM	2	1.3×10^{16}	25.1	0.7%	4.5×10^{35}

Table 2: Unification analysis. M_{GUT} is the geometric mean of the three pairwise crossing scales; τ_p is the dimensional estimate of Proposition 4.

6 Proton lifetime

Proposition 4 (Proton-lifetime estimate). *The dimensional estimate for gauge-boson-mediated $p \rightarrow e^+ \pi^0$,*

$$\tau_p \sim \frac{M_{\text{GUT}}^4}{\alpha_{\text{GUT}}^2 m_p^5}, \quad (11)$$

gives $\tau_p \sim 10^{35-36}$ yr at the MSSM unification point — consistent with the Super-Kamiokande bound $\tau_p > 2.4 \times 10^{34}$ yr [14] — while the SM near-miss scale ($\sim 10^{14-15}$ GeV) would imply $\tau_p \sim 10^{30-31}$ yr, excluded by four orders of magnitude. Proof: by computation; order-of-magnitude only.

The estimate carries $\mathcal{O}(10^{1-2})$ uncertainty from hadronic matrix elements and thresholds; the comparison with experiment is order-of-magnitude. The excluded SM scenario is the classic demise of minimal non-supersymmetric SU(5) [1, 14]. Conversely, LHC limits (gluinos above 2.3 TeV, squarks above 1.85 TeV in simplified models [15]) pressure the $m_{\text{SUSY}} \sim 1$ TeV assumption from below; the dependence of Proposition 3 on m_{SUSY} is logarithmic and is exposed as an interactive parameter in the accompanying application.

7 Limitations and open problems

What would be required for this framework to rise from organized compilation to scientific theory (after [19], §6.4):

Open Problem 1. A single unifying gauge group with the full three-generation dynamics — Yukawa sector, mixing angles, symmetry breaking — derived, not only the one-generation group theory of Proposition 1.

Open Problem 2. Derived inter-sector couplings: matter–gravity, gauge–gravity, and scalar-sector interactions following from one principle.

Open Problem 3. A cure for gravitational non-renormalizability: the two-loop divergence of quantized Einstein gravity [9, 10] must be addressed by a demonstrated mechanism — finiteness (string theory), a non-trivial fixed point (asymptotic safety), or otherwise. The formal loop expansion S_{quantum} of Definition 1 presupposes exactly the consistency that is at issue.

Open Problem 4. Selection in the gravity sector: exactly one UV completion, with the others recovered or excluded in appropriate limits (Remark 1).

Open Problem 5. A novel falsifiable prediction beyond those of the constituent theories: a proton lifetime with controlled hadronic uncertainties, a superpartner mass, or a measurable coupling deviation.

8 Conclusion

This paper presents the honest version of an ambitious project. The composite action of Definition 1 organizes — it does not unify. The quantitative content (Propositions 1–4) is a reproduction of the classic unification analysis, included because any work that speaks about unification must compute what is computable at its level, and made fully reproducible as running code. The distance between this framework and a theory of everything is stated precisely as Open Problems 1–5; none is solved here, and none is solved anywhere. The project’s value is pedagogical and structural: the assembled inputs to the unification problem, the computable parts computed, and the gap to a real theory kept in plain sight.

References

- [1] H. Georgi and S. L. Glashow, *Unity of all elementary-particle forces*, Phys. Rev. Lett. **32**, 438 (1974).
- [2] H. Georgi, H. R. Quinn, and S. Weinberg, *Hierarchy of interactions in unified gauge theories*, Phys. Rev. Lett. **33**, 451 (1974).
- [3] H. Fritzsch and P. Minkowski, *Unified interactions of leptons and hadrons*, Ann. Phys. **93**, 193 (1975).
- [4] S. Dimopoulos and H. Georgi, *Softly broken supersymmetry and $SU(5)$* , Nucl. Phys. B **193**, 150 (1981).
- [5] U. Amaldi, W. de Boer, and H. Fürstenau, *Comparison of grand unified theories with electroweak and strong coupling constants measured at LEP*, Phys. Lett. B **260**, 447 (1991).
- [6] P. Martin and P. Ramond, *Sorting out the minimal supersymmetric standard model*, arXiv:hep-ph/9501244.
- [7] S. P. Martin, *A supersymmetry primer*, arXiv:hep-ph/9709356.
- [8] M. E. Machacek and M. T. Vaughn, *Two-loop renormalization group equations in a general quantum field theory*, Nucl. Phys. B **222**, 83 (1983); **236**, 221 (1984); **249**, 70 (1985).
- [9] M. H. Goroff and A. Sagnotti, *The ultraviolet behavior of Einstein gravity*, Nucl. Phys. B **266**, 709 (1986).
- [10] G. ’t Hooft and M. Veltman, *One-loop divergencies in the theory of gravitation*, Ann. Inst. H. Poincaré A **20**, 69 (1974).
- [11] C. Rovelli, *Quantum Gravity*, Cambridge University Press (2004).
- [12] J. Polchinski, *String Theory*, Vols. 1–2, Cambridge University Press (1998).
- [13] S. Navas *et al.* (Particle Data Group), *Review of particle physics*, Phys. Rev. D **110**, 030001 (2024).
- [14] A. Takenaka *et al.* (Super-Kamiokande), *Search for proton decay via $p \rightarrow e^+\pi^0$ and $p \rightarrow \mu^+\pi^0$ with an enlarged fiducial volume*, Phys. Rev. D **102**, 112011 (2020), arXiv:2010.16098.

- [15] ATLAS Collaboration, *Search for squarks and gluinos in final states with jets and missing transverse momentum at $\sqrt{s} = 13$ TeV (139 fb^{-1})*, JHEP **02**, 143 (2021), arXiv:2010.14293.
- [16] N. Aghanim *et al.* (Planck), *Planck 2018 results VI: Cosmological parameters*, Astron. Astrophys. **641**, A6 (2020), arXiv:1807.06209.
- [17] S. Weinberg, *The cosmological constant problem*, Rev. Mod. Phys. **61**, 1 (1989).
- [18] M. E. Peskin and D. V. Schroeder, *An Introduction to Quantum Field Theory*, Westview Press (1995).
- [19] *Professor Codephreak's GUToE: An Independent Academic Evaluation of an AI-Assisted Theory of Everything* (2026), PDF in the repository root.